

INNOVA JUNIOR COLLEGE

JC 2 PRELIMINARY EXAMINATION 2

in preparation for General Certificate of Education Advanced Level

Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

9647/01

Paper 1 Multiple Choice

21 September 2015

1 hour

Additional Materials: Data Booklet
Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

This document consists of **16** printed pages and **0** blank page.



Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 How many neutrons are present in 0.13 g of ^{13}C ?
[L = the Avogadro constant]

A 0.06 L **B** 0.07 L **C** 0.13 L **D** 0.91 L

- 2 Equimolar amounts of F_2 and BrO_x^- ions react with OH^- ions to produce three products: H_2O , F^- ions and BrO_4^- ions.

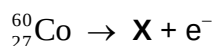
What is the value of x ?

A 1 **B** 2 **C** 3 **D** 5

- 3 Which gas can be most easily liquefied by cooling and applying pressure?

A He **B** CH_4 **C** F_2 **D** NH_3

- 4 Fresh fruits and vegetables are subjected to radiation from the radioactive decomposition of $^{60}_{27}\text{Co}$ to eliminate harmful microorganisms to prolong their shelf-life. During the decomposition, a neutron in the nucleus decomposes to form a proton, which is retained in the nucleus and an electron which is expelled from the atom. No other particle is produced.



Which row in the table correctly describes the nuclear make-up of $^{60}_{27}\text{Co}$ and **X**?

	$^{60}_{27}\text{Co}$	X	
	Number of neutrons	Number of protons	Number of neutrons
A	33	26	31
B	33	28	32
C	60	26	34
D	60	28	32

- 5 In which pair of molecules are the values of the bond angles the closest?

A BF_3 and NH_3
B C_2H_4 and BF_3
C H_2O and C_2H_4
D CH_4 and H_2O

6 In which change would only van der Waals' forces have to be overcome?

- A evaporation of ethanol
- B melting of ice
- C** melting of solid carbon dioxide
- D solidification of butane

7 Hess's Law can be used to calculate the average C–H bond energy in methane.

$\Delta H_{\text{at}}^{\ominus}$ = standard enthalpy change of atomisation

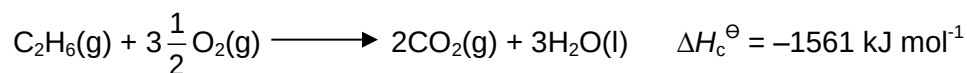
$\Delta H_{\text{f}}^{\ominus}$ = standard enthalpy change of formation

$\Delta H_{\text{c}}^{\ominus}$ = standard enthalpy change of combustion

Which of the following information is needed in order to perform the calculation?

- A $\Delta H_{\text{f}}^{\ominus}(\text{CH}_4)$ only
- B** $\Delta H_{\text{at}}^{\ominus}(\text{C})$, $\Delta H_{\text{at}}^{\ominus}(\text{H})$, $\Delta H_{\text{f}}^{\ominus}(\text{CH}_4)$
- C $\Delta H_{\text{c}}^{\ominus}(\text{C})$, $\Delta H_{\text{c}}^{\ominus}(\text{H}_2)$, $\Delta H_{\text{c}}^{\ominus}(\text{CH}_4)$
- D $\Delta H_{\text{c}}^{\ominus}(\text{C})$, $\Delta H_{\text{c}}^{\ominus}(\text{H}_2)$, $\Delta H_{\text{f}}^{\ominus}(\text{CH}_4)$

8 Ethane undergoes combustion as follows:



Given the following data, calculate the standard enthalpy change of vapourisation of $\text{H}_2\text{O}(\text{l})$.

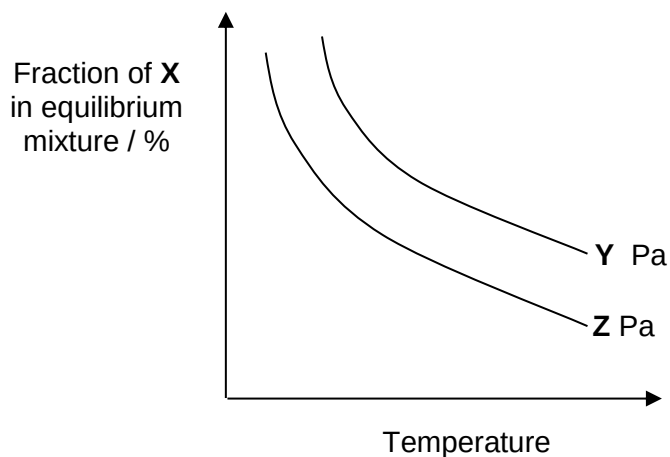
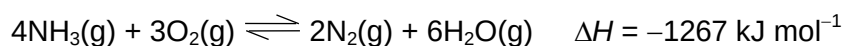
$\Delta H_{\text{c}}^{\ominus}$ of $\text{C}(\text{s}) = -394 \text{ kJ mol}^{-1}$

$\Delta H_{\text{f}}^{\ominus}$ of $\text{C}_2\text{H}_6(\text{g}) = -85 \text{ kJ mol}^{-1}$

$\Delta H_{\text{f}}^{\ominus}$ of $\text{H}_2\text{O}(\text{g}) = -243 \text{ kJ mol}^{-1}$

- | | |
|------------------------------------|-------------------------------------|
| A $+43 \text{ kJ mol}^{-1}$ | C $+129 \text{ kJ mol}^{-1}$ |
| B -43 kJ mol^{-1} | D -129 kJ mol^{-1} |

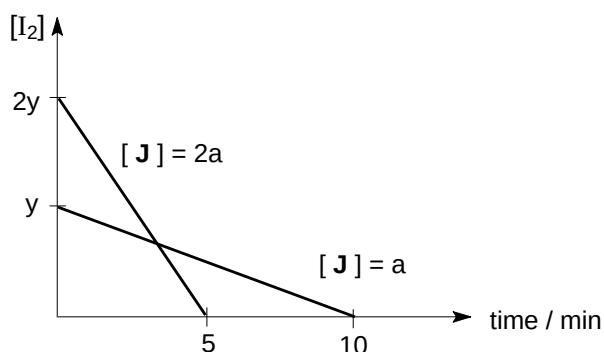
- 9 The graph below shows how the fraction of a substance, **X** represented by one of the following compounds in the equilibrium mixture shown below varies with temperature at pressures of **Y** Pa and **Z** Pa.



Identify **X** and the correct relative magnitudes of **Y** and **Z**.

	X	Pressure
A	N_2	$\text{Z} > \text{Y}$
B	O_2	$\text{Y} > \text{Z}$
C	H_2O	$\text{Y} > \text{Z}$
D	NH_3	$\text{Z} > \text{Y}$

- 10 The kinetics of the reaction between iodine and compound **J** is investigated.



What conclusions can be drawn from the graphs?

- A** The reaction is second order with respect to compound **J** because rate of reaction increases by four times when its concentration is increased by two times.
- B** Both iodine and compound **J** are involved in the rate determining step.
- C** The reaction is first order with respect to iodine because half-life is constant.
- D** The overall order of the reaction is 1.

- 11** A radioactive element has 2 isotopes, **G** and **H**, with half-lives of 2 days and 4 days respectively. An experiment starts with 4 times as many atoms of **G** as of **H**.

Radioactive decay is a first-order reaction.

How long will it be before the number of atoms of **G** left equals the number of atoms of **H** left?

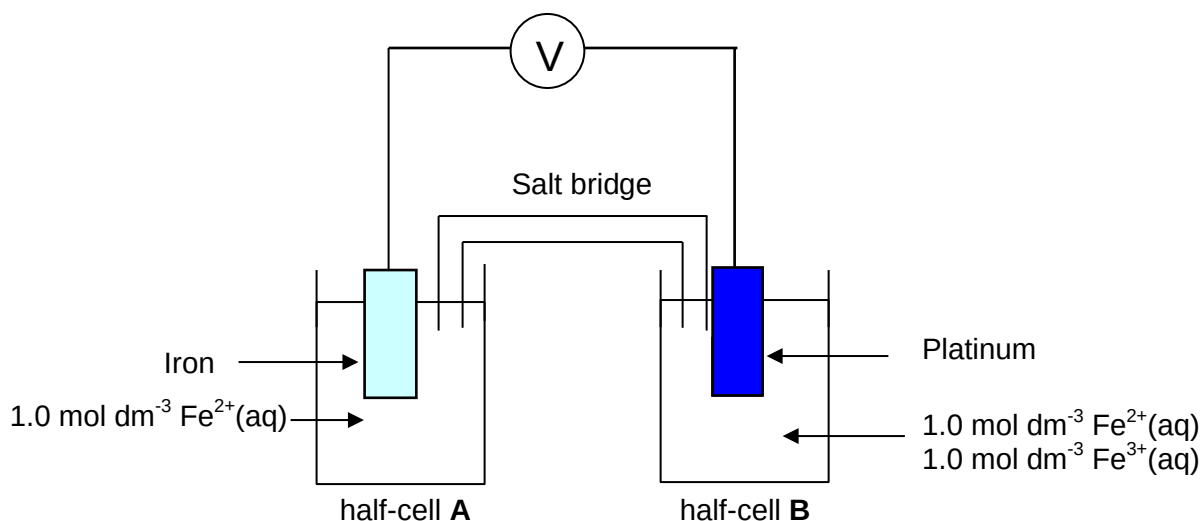
- A** 8 days
B 32 days
C 16 days
D 64 days

- 12** Which species is dominant when equal volumes of aqueous solutions of 0.1 mol dm^{-3} H_2SO_4 , 0.1 mol dm^{-3} NaOH and 0.1 mol dm^{-3} $\text{H}_2\text{NCH}(\text{CH}_3)\text{COOH}$ are mixed?

- A** $\text{H}_2\text{NCH}(\text{CH}_3)\text{COOH}$ **C** $\text{H}_3\text{N}^+\text{CH}(\text{CH}_3)\text{COO}^-$
B $\text{H}_2\text{NCH}(\text{CH}_3)\text{COO}^-$ **D** $\text{H}_3\text{N}^+\text{CH}(\text{CH}_3)\text{COOH}$

- 13** *Use of the Data Booklet is relevant to this question.*

The cell shown in the diagram below is set up, and the initial $E^\ominus_{\text{cell}}$ value was taken.



In experiment 1, some CN^- ions was added to half-cell **A**, before the $E^\ominus$ value was measured.

In experiment 2, excess CN^- ions was added to half-cell **B**, before the $E^\ominus$ value was measured.

How will the $E^\ominus_{\text{cell}}$ value in experiment 1 and 2 differ from the initial $E^\ominus_{\text{cell}}$ value?

	experiment 1	experiment 2
A	less positive	more positive
B	less positive	less positive
C	more positive	less positive
D	more positive	more positive

- 14 Which of the following elements has an oxide with a giant structure and a chloride which is readily hydrolysed?
- A aluminium
 B carbon
 C phosphorus
D silicon

- 15 For the elements in the third period of the Periodic Table, which property decreases consistently from sodium to chloride?
- A electrical conductivity
 B ionisation energy
 C melting point
D radius of the atom

- 16 The carbonates of Group II decompose according to the following equation.



For this reaction, ΔH increases on descending down the group.

carbonate	Mg	Ca	Sr	Ba
$\Delta H / \text{kJ mol}^{-1}$	+101	+178	+235	+269

Which property best explains this trend?

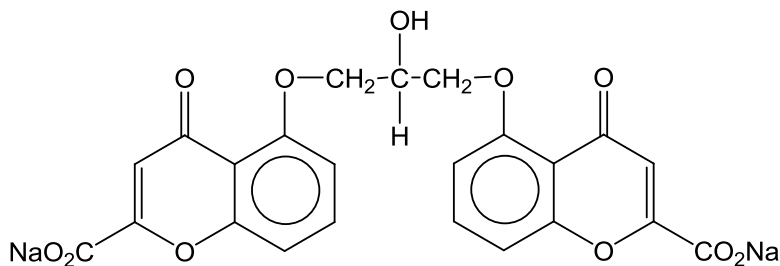
- A** the ionic radius of the metal ion
 B the ionisation energy of the metal
 C the proton number of the metal
 D the thermal stability of the oxide
- 17 Which of the following statements about the trends in the properties of the halogens down the group is correct?
- A The volatility increases.
 B The electronegativity increases.
 C The reactivity as oxidising agents increases.
D The enthalpy change of reaction with hydrogen becomes more endothermic.

- 18

Which of the following classifications for the reactions is **incorrect**?

	reaction	acid-base	ligand exchange	precipitation	redox
A	$2\text{FeCl}_2 + \text{Cl}_2 \longrightarrow 2\text{FeCl}_3$				✓
B	$[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 4\text{NH}_3 \longrightarrow [\text{Cu}(\text{NH}_3)_4]^{2+} + 6\text{H}_2\text{O}$	✓	✓		
C	$[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 4\text{HCl} \longrightarrow [\text{CuCl}_4]^{2-} + 4\text{H}^+ + 6\text{H}_2\text{O}$		✓		
D	$[\text{Fe}(\text{H}_2\text{O})_6]^{3+} + 3\text{OH}^- \longrightarrow \text{Fe}(\text{OH})_3 + 6\text{H}_2\text{O}$	✓		✓	

- 19



How many chiral centres are there in the molecule?

- | | | | |
|----------|---|----------|---|
| A | 0 | C | 1 |
| B | 2 | D | 3 |

- 20

Which bond is not present in 1-penten-3-yne, $\text{CH}_3\text{C}\equiv\text{CCH}=\text{CH}_2$?

- A** a π bond formed by 2p-2p overlap
B a σ bond formed by 1s-2sp overlap
C a σ bond formed by 2sp-2sp² overlap
D a σ bond formed by 2sp-2sp³ overlap

- 21** In the free radical substitution of 2-methylbutane with bromine, a mixture of mono-brominated compounds were obtained.

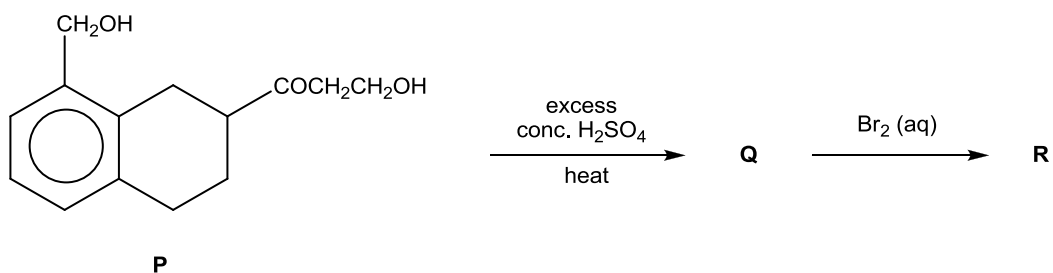
What is the statistical ratio of the two compounds with the highest yield?

- A 1 : 1
- B** 1 : 2
- C 1 : 3
- D 1 : 6

- 22** In which of the following reactions is the inorganic reagent acting as a nucleophile?

- A $\text{C}_6\text{H}_6 + \text{HNO}_3 \longrightarrow \text{C}_6\text{H}_5\text{NO}_2 + \text{H}_2\text{O}$
- B $\text{CH}_3\text{CH}=\text{CH}_2 + \text{Br}_2 \longrightarrow \text{CH}_3\text{CHBrCH}_2\text{Br}$
- C $\text{CH}_3\text{CH}_2\text{NH}_2 + \text{HCl} \longrightarrow \text{CH}_3\text{CH}_2\text{NH}_3\text{Cl}$
- D** $\text{CH}_3\text{CH}_2\text{Br} + \text{NaOH} \longrightarrow \text{CH}_3\text{CH}_2\text{OH} + \text{NaBr}$

23 Compound **P** is used in the following synthesis.



What is the likely structure of the final product **R**?

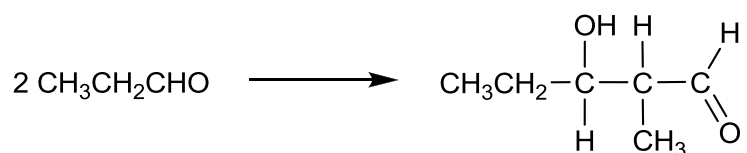
- A**
-
- B**
-
- C**
-
- D**
-

24 A bromine-containing organic compound, **T**, undergoes an elimination reaction when treated with hot ethanolic sodium hydroxide solution.

What is **T**?

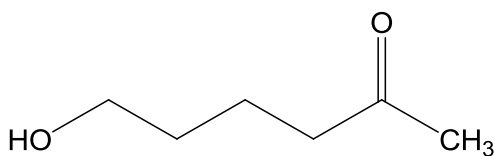
- A** CH_3Br
- B** C_2Br_6
- C** $\text{Br}_2\text{C}=\text{CH}_2$
- D** $(\text{CH}_3)_2\text{C}=\text{CBr}_2$

- 25** The *aldol condensation* reaction occurs between carbonyl compounds. An example of the reaction is given below:

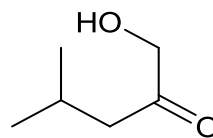


Which of the following compounds is the product of an aldol condensation reaction between 2 identical carbonyl compounds?

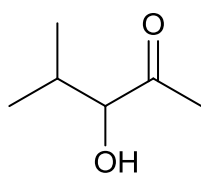
A



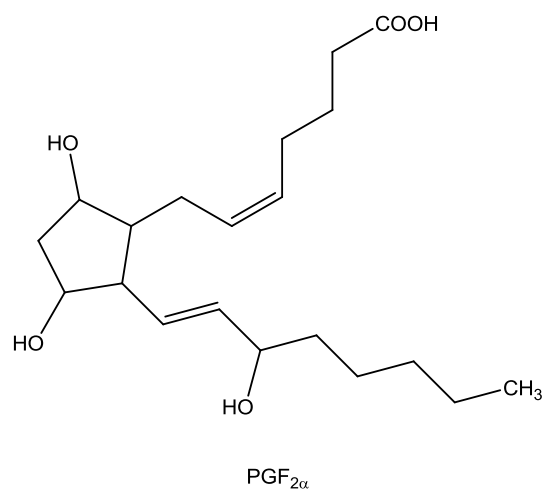
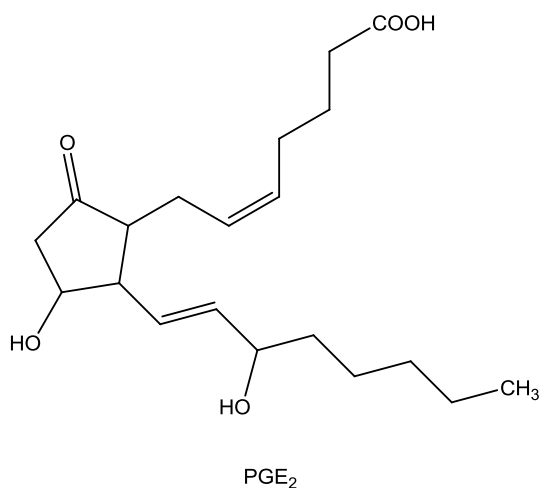
C



B

**D**

- 26** PGE₂ and PGF_{2α} are two prostaglandins, both with pharmacological activity.



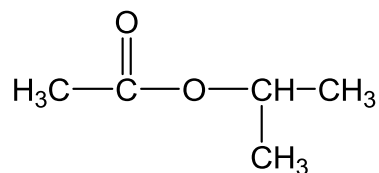
Which reagent will convert PGE₂ into PGF_{2α} efficiently?

A H₂/NiC LiAlH₄B H₂O/H⁺**D** NaBH₄

- 27** Which sequence shows the decreasing ease of hydrolysis?

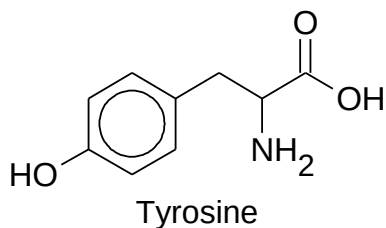
A C₆H₅Br > C₆H₅CH₂Cl > CH₃CH₂COC₂H₅ > CH₃CH₂I**B** CH₃CH₂COC₂H₅ > CH₃CH₂I > C₆H₅CH₂Br > C₆H₅CH₂ClC C₆H₅Cl > CH₃COCH₂Cl > C₆H₅CH₂Br > CH₃CH₂ID C₆H₅Br > CH₃CH₂Cl > C₆H₅CH₂I > CH₃CH₂COC₂H₅

28 Which one of the statements about the following ester is correct?



- A It is a chain isomer of pentanoic acid.
- B It is a functional group isomer of ethyl propanoate.
- C** It can be hydrolysed to produce an alcohol that can also be formed by the acid-catalysed hydration of propene.
- D It can be hydrolysed to produce an alcohol that is resistant to oxidation by acidified potassium dichromate (VI).

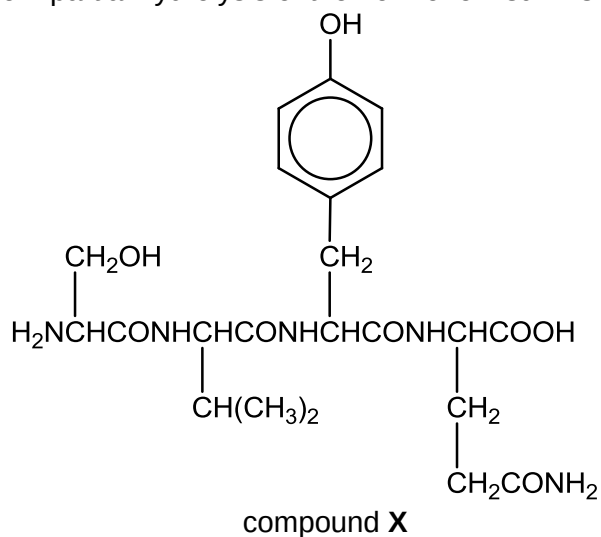
29 Tyrosine is one of the 20 amino acids that is used by cells to synthesise proteins, and was first discovered in cheese.



Which of the following types of reaction will Tyrosine **not** undergo?

- | | |
|-----------------------------|------------------------------|
| A reduction | C acid hydrolysis |
| B nucleophilic substitution | D electrophilic substitution |

30 Compound **X** formed from partial hydrolysis of the hormone insulin is shown below.



Which of the following statements is **incorrect**?

- A** Prolonged heating of compound **X** with dilute HCl produces 4 carbon-containing products.
- B** Prolonged heating of compound **X** with dilute NaOH liberates an alkaline gas.
- C** 6 moles of NaOH is needed for complete reaction with 1 mole of compound **X**.
- D** 2 moles of Na_2CO_3 is needed for complete reaction with 1 mole of compound **X**.

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

31 Two ions formed by the isotopes of iron have chemical formulae shown below:



Which of the following statements about these two ions are **incorrect**?

- 1** The electronic configuration of both Fe^{2+} is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$.
- 2** ${}_{26}^{56}\text{Fe}^{2+}$ gives a greater angle of deflection than ${}_{26}^{54}\text{Fe}^{2+}$ in an electric field.
- 3** The total number of protons and electrons in each of the Fe^{2+} ions is smaller than the nucleon number of the respective ion.

32 The definitions of many chemical terms can be illustrated by chemical equations.

Which terms can be illustrated by an equation that shows the formation of a positive ion?

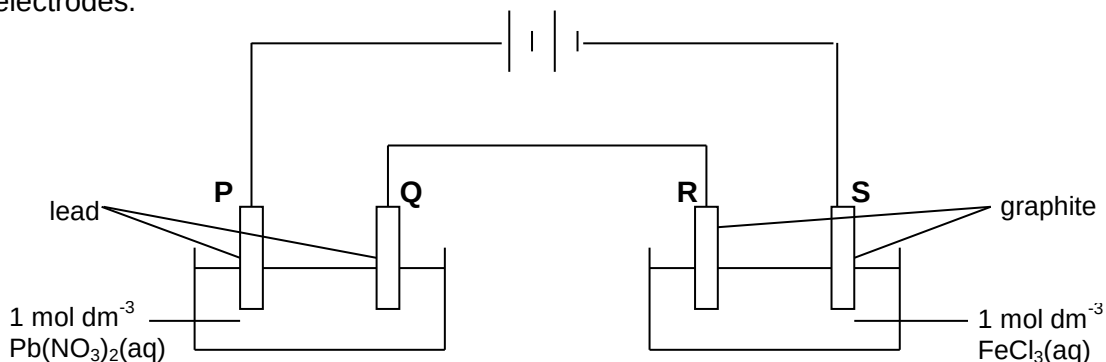
- 1** first ionisation energy
- 2** heterolytic fission
- 3** electron affinity

33 Which of the following mixtures would constitute a buffer solution?

- 1** 1 mol of H_2PO_4^- and 1 mol of HPO_4^{2-}
- 2** 1 mol of HCl and 2 mol of NH_3
- 3** 1 mol of NaOH and 2 mol of 1,4-benzenedicarboxylic acid

34 Use of the Data Booklet is relevant to this question.

Two electrolytic cells are connected in series as shown in the diagram where P, Q, R and S are electrodes.



Which statements about the products formed at each electrode are correct?

- 1 Fe^{2+} is formed at electrode S.
- 2 H_2 is formed at electrode Q.
- 3 O_2 is formed at electrode P.

35 The Group II metals have higher melting points than the Group I metals.

Which factors could contribute towards the higher melting points?

- 1 There are smaller interatomic distances in the metallic lattices of the Group II metals.
- 2 Two valence electrons are available from each Group II metal atom for bonding the atom into the metallic lattice.
- 3 Group II metals have the higher first ionisation energy.

36 Use of the Data Booklet is relevant to this question.

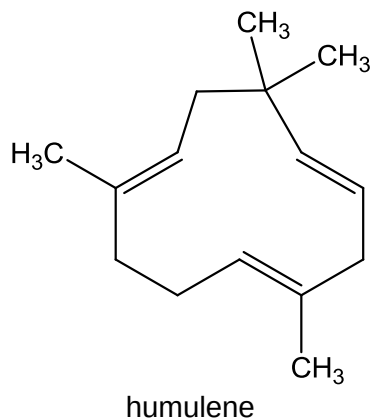
Which pairs of reagents when reacted together would produce a change in the oxidation numbers of the metal atoms?

- 1 aqueous potassium manganate(VII) and acidified aqueous iron(II) sulfate
- 2 aqueous acidified lead(IV) chloride and aqueous vanadium(III) nitrate
- 3 aqueous ammonia and an aqueous suspension of silver chloride

37 Which reactions yield a carbon compound incorporating deuterium, D? [D = ^2H]

- 1 CH_3COCl $\xrightarrow[\text{D}_2\text{O}]{\text{NaOD}}$
- 2 $\text{CH}_3\text{CH}_2\text{CN}$ $\xrightarrow[\text{D}_2\text{O}]{\text{NaOD}}$
- 3 $(\text{CH}_3)_3\text{COH}$ $\xrightarrow[\text{heat}]{\text{conc. D}_2\text{SO}_4}$

- 38** Humulene can be extracted from carnation flowers.



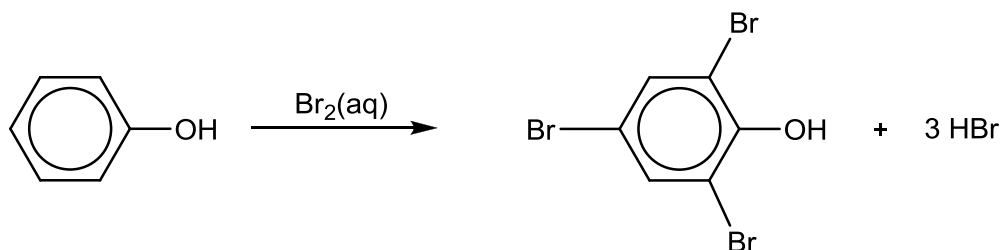
Which products are obtained from the reaction of humulene with hot acidified concentrated KMnO_4 ?

- 1** $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CO}_2\text{H}$
- 2** $\text{CH}_3\text{COCH}_2\text{CO}_2\text{H}$
- 3** $\text{HO}_2\text{CCH}_2\text{C}(\text{CH}_3)_2\text{CO}_2\text{H}$

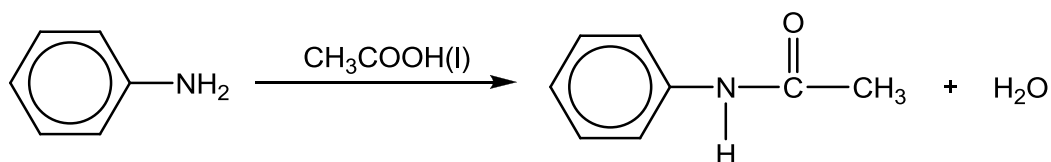
- 39** Three different transformations are given.

Which transformations are correct?

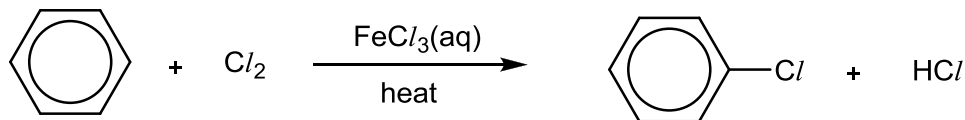
1



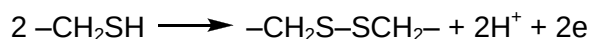
2



3



- 40** Keratin, a family of fibrous protein, is a key structural component in hair. The curliness of the hair is determined by the number of disulfide bonds that can be formed between R groups ($-\text{CH}_2\text{SH}$) of the cysteine residues of adjacent keratin molecules. The formation of disulfide bonds can be represented by the following equation:



With more disulfide bonds present, the hair is curlier.

In hair rebonding, heat or chemical treatment may be applied to curly hair to straighten the hair.

Which of the following are true?

- 1** For hair rebonding using heat treatment, the disulfide bonds are overcome.
- 2** An oxidising agent may be used in the chemical treatment to straighten the hair.
- 3** The primary structure of keratin is disrupted during hair rebonding.

Final Answer

1	B	6	C	11	A	16	A	21	B	26	D	31	B	36	B
2	C	7	B	12	D	17	D	22	D	27	B	32	B	37	D
3	D	8	A	13	C	18	B	23	D	28	C	33	A	38	A
4	B	9	A	14	D	19	A	24	C	29	C	34	D	39	D
5	B	10	A	15	D	20	B	25	D	30	D	35	B	40	D